Contents

1 A proof sketch of the fact that there is no quintic formula **1**

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Theorem. There is no finite formula using a, b, c, d, e, fixed complex constants, addition, multiplication, division, and taking nth roots, that returns a root of $x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0$ for every choice of complex coefficients a, b, c, d, e.

We will only give a proof sketch, since the worries about paths going through branch points are evil little monster abominations that make what would otherwise be a super beautiful argument into a rabbithole of technical justification poop.

Proposition 1. There is no general formula returning a root of every quadratic equation with complex coefficients

Proof. The main proof builds on the idea we are about to use. Suppose there were such a formula. Then it would give a continuous single-valued function from the coefficients to the roots that is single valued and therefore picks a root whenever its denominators are non-zero. Since it is assumed to work for every quadratic, it is defined throughout the loop used below. Since the hypothetical formula uses only continuous operations and division, it is continuous wherever all its denominators are nonzero. Because it is assumed to be defined for every quadratic, those denominators must remain nonzero throughout the loop $x^2 = e^{i\theta}$. Thus it selects a root continuously along the entire loop.

Now suppose this hypothetical formula lets us find the root 1 of $x^2 - 1 = 0$, or the root -1 as it would be the same idea. Now write the equation above as $x^2 = 1$. Now we know that if we have $x^2 = e^{i\theta}$ the roots are $x = \pm e^{\frac{i\theta}{2}}$. Now lets look at how the root 1 changes as we vary θ continuously from 0 to 2π. What would end up happening is the following:

$x^2 = e^{i0}$	Roots: $1,\ -1$
$x^2 = e^{i0.1}$	Roots: $e^{i0.05},\ -e^{i0.05}$
...	...
$x^2 = e^{i2\pi}$	Roots: $e^{i\pi} = -1,\ -e^{i\pi} = 1$

The problem is, now, that the root returned by the formula must be the one on the left in the table above – it cannot change at any point in the process to being the one on the right because otherwise it would not be continuous. Therefore our formula would return both 1 and -1 for $x^2 = 1$ but our formula is single valued. Contradiction.

□

A radicand is an expression inside a radical, for example the radicand of $\sqrt{1+z}$ is $1 + z$. Start at some coefficients and choose a value for each radical, working from the inner most ones. For example, in $\sqrt{1+\sqrt{z}}$ near $z = 4$ we could start with $\sqrt{4} = 2$ and then choose the outer square root to be $\sqrt{3}$. Small changes in z then give unique nearby choices of both values. We will always choose starting points where each radicand and denominator is non-zero

We now prove a proposition about cubics and quartics to make it make sense what we will do with the quintic as it would not make sense if we straight up did it. Replace any expressions that are identically 0 (like $\sqrt{x-x}$) with 0.

Proposition 2. Any general formula for a cubic ($x^3 + ax^2 + bx + c = 0$) using only $+$, *, /, the complex numbers, and n'th roots must have nesting depth at least 2.

A formula with nesting depth 2 exists but it is very complicated (image below) so generally numerical methods are more practical.

$$x = \sqrt[3]{\left(\frac{-b^3}{27a^3}+\frac{bc}{6a^2}-\frac{d}{2a}\right)+\sqrt{\left(\frac{-b^3}{27a^3}+\frac{bc}{6a^2}-\frac{d}{2a}\right)^2+\left(\frac{c}{3a}-\frac{b^2}{9a^2}\right)^3}} + \sqrt[3]{\left(\frac{-b^3}{27a^3}+\frac{bc}{6a^2}-\frac{d}{2a}\right)-\sqrt{\left(\frac{-b^3}{27a^3}+\frac{bc}{6a^2}-\frac{d}{2a}\right)^2+\left(\frac{c}{3a}-\frac{b^2}{9a^2}\right)^3}} - \frac{b}{3a}.$$

Figure 1

Figure 1 showsthe cubic formula for $ax^3 + bx^2 + cx + d = 0$ with $a \neq 0$. If two cube root terms are U, V choose them so that $UV = \frac{b^2}{9a^2} - \frac{c}{3a}$. Which polynomial root we get depends on this compatible pair of cube roots. Changing the sign of the square root merely interchanges the two terms when their cube-root values are interchanged with them.

Sketch of proof of proposition 2. Suppose we had a formula with nesting depth 1 – since 0 can be ruled out by the quadratic argument. Then pick a root in the formula and call the stuff inside it X. X is continuous and single valued, since it just uses the basic math operations, and it is continuous along any path we use to permute the roots (this is where it becomes a sketch). Now take the roots and label them r_1, r_2 and r_3 and suppose that at this particular point our hypothetical formula outputs r_1. Now swap r_1 and r_2 continuously such that X avoids 0 and infinity. When we do this, X will go back to where it started – X is a function symmetric in the roots and we merely reordered the roots. If X is inside an n'th root, and wrapped around the origin m times, then $\sqrt[n]{X}$ will move an angle $2\pi\frac{m}{n}$ anticlockwise. This is why we needed to make sure we do not touch the origin – we either wrap around it some times or we do not, and we do not blow up to infinity. Now swap r_2 and r_3. This will cause X to possibly or possibly not wrap around the origin. Lets say it wraps p times anticlockwise, then $\sqrt[n]{X}$ will have moved a total angle of $2\pi\frac{m+p}{n}$ anticlockwise. Now permute the roots in cycle notation by doing the permutation (2 3)(1 2)(2 3)(1 2) and always using the same paths. This is the cycle (1 2 3) so it leaves the roots different, but $\sqrt[n]{X}$will have moved a total angle of $2\pi\frac{m+p-p-m}{n} = 0$ anticlockwise (i mean changed that much in argument by the end, not necessarily to have moved in a circle!). This is a problem because all of the roots in our hypothetical equation, and thus the value of the equation, will be the same. The same winding cancellation holds for every radical in our hypothetical formula, each with its own winding numbers. Therefore every radical value, and hence the value of the whole formula, returns to its starting value. We supposed it returned r_1, it never switched to returning another root since the roots moved continuously and never crossed and the function from the roots to the coefficients to X is continuous, and the roots never crossed to allow it to jump. So our formula returns r_1, which moved, but also returns the same thing that it returned at the start since it ended up back where it started. Contradiction.

Proposition 3. Any general formula for a quartic ($x^4 + ax^3 + bx^2 + cx + d = 0$) using only +, *, /, the complex numbers, and n'th roots must have nesting depth at least 3.

There is a formula with nesting depth 3 but it literally does not fit on a single line in latex, and I'm not intending to make a Fermat reference. Obviously, numerical methods are more practical. This is the exact form for the larger real root of $x^4 - x - 1 = 0$:

Figure 2 shows an image of the exact form from wolframalpha

$$x = \frac{1}{2}\sqrt{\frac{\sqrt[3]{\frac{1}{2}\left(9+\sqrt{849}\right)}}{3^{2/3}} - 4\sqrt[3]{\frac{2}{3\left(9+\sqrt{849}\right)}}} + \frac{1}{2}\sqrt{4\sqrt[3]{\frac{2}{3\left(9+\sqrt{849}\right)}} - \frac{\sqrt[3]{\frac{1}{2}\left(9+\sqrt{849}\right)}}{3^{2/3}} + \frac{2}{\sqrt{\frac{\sqrt[3]{\frac{1}{2}\left(9+\sqrt{849}\right)}}{3^{2/3}} - 4\sqrt[3]{\frac{2}{3\left(9+\sqrt{849}\right)}}}}}}$$

Figure 2

This actually has nesting depth 4 but we will prove at least 3 is needed.

Sketch of proof of proposition 3. The idea we did for proposition 2 (Path A, B, A backwards then B backwards) is called a commutator. This is commonly used on the Rubik's cube. We will exploit this idea more.

So we can do a commutator to rule out nesting depth 1 like we did for the cubic. This leaves any roots without roots

inside them unchanged, but we do not know what it does to everything inside the root with a nesting depth 1 expression inside it – we just know that after a commutator the stuff inside the root is back to where it started, but it could have wrapped around the origin in the process. But now lets do a commutator of commutators and track the nesting depth 1 expressions. We know a commutator of swaps containing exactly one shared element is a 3-cycle. A commutator of three cycles: $(1\ 2\ 3)(2\ 3\ 4)(1\ 3\ 2)(2\ 4\ 3) = (1\ 4)(2\ 3)$. This will make, for the same reason, the number of total times the nesting depth 1 expressions wrap around the origin be 0, therefore any nesting depth 2 expression will be back where it started and we have the same contradiction.

Sketch of proof of the main theorem. This is easy once you have understood the concepts used above. We just need to rule out any finite nesting depth. But we can have an infinite chain of commutators of commutators of commutators of ... This is because a certain commutator of three cycles has the property $(1\ 2\ 3)\,(1\ 4\ 5)\,(1\ 3\ 2)\,(1\ 5\ 4) = (1\ 2\ 4)$, which is another 3-cycle. Therefore we can make a chain as long as we want where at each step we just have a 3-cycle, and we can pick this in a way that it moves the root our hypothetical formula outputs. We can therefore get the same contradiction. This completes the proof.